

10.18 Consider the data file *mroz* on working wives. Use the 428 observations on married women who participate in the labor force. In this exercise, we examine the effectiveness of a parent's college education as an instrumental variable.

- a. Create two new variables. *MOTHERCOLL* is a dummy variable equaling one if *MOTHEREDUC* > 12, zero otherwise. Similarly, *FATHERCOLL* equals one if *FATHEREDUC* > 12 and zero otherwise. What percentage of parents have some college education in this sample?

About 12.14953% of mothers and 11.68224% of fathers have some college education in this sample.

- b. Find the correlations between *EDUC*, *MOTHERCOLL*, and *FATHERCOLL*. Are the magnitudes of these correlations important? Can you make a logical argument why *MOTHERCOLL* and *FATHERCOLL* might be better instruments than *MOTHEREDUC* and *FATHEREDUC*?

$\text{corr}(\text{EDUC}, \text{MOTHERCOLL}) \approx 0.3597$

$\text{corr}(\text{EDUC}, \text{FATHERCOLL}) \approx 0.3985$

$\text{corr}(\text{MOTHERCOLL}, \text{FATHERCOLL}) \approx 0.3546$

These correlations are moderately large and positive.

Therefore, both *MOTHERCOLL* and *FATHERCOLL* are correlated with *EDUC*, which suggests that they satisfy the relevance condition for instrumental variables.

A logical argument for using *MOTHERCOLL* and *FATHERCOLL* instead of *MOTHEREDUC* and *FATHEREDUC* is that the effect of parents' education on a child's education may be nonlinear.

Whether a parent attended college may matter more than each additional year of parental schooling.

Therefore, the dummy variables may better capture the important educational background of the parents

- c. Estimate the wage equation in Example 10.5 using *MOTHERCOLL* as the instrumental variable. What is the 95% interval estimate for the coefficient of *EDUC*?

The wage equation:

$$\ln(WAGE) = \beta_1 + \beta_2 EXPER + \beta_3 EXPER^2 + \beta_4 EDUC + e$$

The first-stage equation:

$$EDUC = \gamma_1 + \gamma_2 EXPER + \gamma_3 EXPER^2 + \theta_1 MOTHERCOLL + v$$

The IV estimate of the coefficient on *EDUC* is 0.0760, with a standard error of 0.0394

The 95% interval estimate for the coefficient of *EDUC* is approximately [-0.0012, 0.1533].

- d. For the problem in part (c), estimate the first-stage equation. What is the value of the *F*-test statistic for the hypothesis that *MOTHERCOLL* has no effect on *EDUC*? Is *MOTHERCOLL* a strong instrument?

$$F = t^2$$

$$F = 63.563 = 7.973^2, \quad p\text{-value} = 1.455 \times 10^{-10}$$

Reject H_0 : *MOTHERCOLL* has no effect on *EDUC*

Yes, *MOTHERCOLL* is a strong instrument. ($\because F > 10$)

- e. Estimate the wage equation in Example 10.5 using *MOTHERCOLL* and *FATHERCOLL* as the instrumental variables. What is the 95% interval estimate for the coefficient of *EDUC*? Is it narrower or wider than the one in part (c)?

The wage equation:

$$\ln(WAGE) = \beta_1 + \beta_2 EXPER + \beta_3 EXPER^2 + \beta_4 EDUC + e$$

The first-stage equation:

$$EDUC = \gamma_1 + \gamma_2 EXPER + \gamma_3 EXPER^2 + \theta_1 MOTHERCOLL + \theta_2 FATHERCOLL + v$$

The IV estimate of the coefficient on *EDUC* is 0.0878, with a standard error of 0.0308

The 95% interval estimate for the coefficient of *EDUC* is approximately [0.0275, 0.1482].

- f. For the problem in part (e), estimate the first-stage equation. Test the joint significance of *MOTHERCOLL* and *FATHERCOLL*. Do these instruments seem adequately strong?
- g. For the IV estimation in part (e), test the validity of the surplus instrument. What do you conclude?

f. $F = 56.963$, $p\text{-value} < 2.2 \times 10^{-16}$

Reject $H_0: \theta_1 = 0, \theta_2 = 0$

Yes, these instruments seem adequately strong.

g. H_0 : The instruments are valid

① H_A : The instruments are invalid

② $\alpha = 0.05$

③ Sargan test

④ Sargan statistic = 0.238

⑤ $p\text{-value} = 0.626 > 0.05$, do not reject H_0

There is no evidence that the surplus instrument is invalid.